

Engineering Ethics - Ethical Theories

Ethics is that branch of philosophy that deals with morality. An engineer with ethics is a person who is expected to possess the moral integrity with rich ethical values. The ethics are mainly divided into two categories depending upon the morality of humanity. They are –

Consequential Ethics

The Consequential ethics are values the outcome of which determine the morality behind a particular action. A lie which saves a life, comes under this.

Non-consequential Ethics

The non-consequential ethics are values followed where the source of morality comes from the standard values. The moral law which states that a lie is a lie, and shouldn't be done, though it ends in a good deed can be taken as an example of non-consequential ethics.

Types of Ethical Theories

Depending upon the ethics a person is intended to follow, four theories were postulated by four different philosophers. These theories help to create the fundamentals of obligation suitable and applicable to professional and personal conduct of a person in his everyday life.



Let us discuss each theory in detail.

Golden Mean

The Golden Mean ethical theory was proposed by **Aristotle**. According to this theory, the solution to a problem is found by analyzing the reason and the logic. A **Mean value of solution** which will be between the extremes of excess and deficiency.

For example, the solution to the problem of environment pollution is neither by avoiding industrialization and civilization, nor by neglecting the environment completely. A mean solution that will work towards controlling the pollution and protecting the environment will also help.

Problem in Application

The application of this theory varies from one person to another with their powers of reasoning and the difficulty in applying the theory to ethical problems.

What is Golden Mean?

The Golden Mean virtue can be understood as the virtue of reaching a proper balance between extremes in conduct, emotion, desire and attitude. This theory phrased by Aristotle states that virtues are tendencies to find the **golden mean** between the extremes of too much (excess) and too little (deficiency) with regard to particular aspects of our lives.

The most important virtue is **practical wisdom**, i.e., morally good judgment, which enables one to discern the mean for all the other virtues. There are internal goods such as products, activities and experiences should never clash with the external goods such as money, power self-esteem and prestige. The standards of excellence enable internal goods to be achieved. The external goods when extremely concerned, though by individuals or by organizations, threaten the internal goods.

Rights-based Ethical Theory

The Rights based ethical theory was proposed by **John Locke**. According to this theory, the solution to a problem is by realizing that every person has a right to live. Live and let live is the philosophy behind this theory. The rights of a person towards life, health, liberty, possession, etc. are taken care of under this theory.

For example, any action in terms of Capital punishment, Jails, Income taxes and Medical charges etc. come under this category.

Problem in Application

One rights of a person may be in conflict with rights of the other.

What does it mean?

Rights-based ethics is the recognition of human dignity at its most basic form. The ethics refer to the basic human rights whether they are positive or negative. Everyone has a right to live, liberty and the pursuit of happiness. Beauchamp and Childress, authors and ethical theorists, have defined the term "right" to be a "justified claim that individuals and groups can make upon other individuals or upon society; to have a right is to be in a position to determine by one's choices, what others should do or need not do."

The natural law states that human laws are defined by morality and not by some authority. This law is derived from the belief that human morality comes from nature. Any action done by a person that would prevent a fellow being from living a good and happy life, is considered immoral or unnatural. Any law should have some morals. Moral duty is the obligation to act based on ethical beliefs.

Duty-based Ethical Theory

The duty-based ethical theory was proposed by **Immanuel Kant**. According to this theory, every person has a duty to follow which is accepted universally, with no exceptions.

An example of this can be expecting all to be honest, kind, generous and peaceful.

Problem in Application

The universal application of this theory can be misleading.

What are these ethics?

Kant observed that everyone is bound to follow some moral laws. It is the choice we make to be morally sound though we have chances to do anything. This theory can also be called as **Deontological theory** or the **Absolutist theory**. According to this, it is our duty to obey the categorical imperative rules. To have good will, is to perform ones duty for the sake of duty and for no other reason.

The categorical imperative law states that Act only according to that maxim by which you can at the same time will that it should become a universal law.

There are **four virtues** that come under this law, which have to be discussed here.

Prudence

The quality of prudence states that every individual has a life that should be respected and every individual has duties which should be done without any exception. One should always be cautious to perform one's duties.

Temperance

Temperance is the voluntary self-restraint from the attractions. The temptations that might lead to the violation of duties and ethics have to be restrained. No false promises are to be made as they contradict the principles of duties.

Fortitude

Fortitude is the sense of having tolerance. No perfection can be maintained if happiness alone is sought and no happiness is achieved if perfection alone is sought. Both may or may not go with each other.

Justice

Every individual is a human being with a set of intrinsic values and morals. Truth and fairness are the aspects one should always bear in mind. People should be treated as separate individuals but never as a mere means of existence.

A free will and a will under moral laws are one and the same. We are free only when we act in accordance with our own best natures, while we are slaves whenever we are under the rule of our passions and wills. There should be a universally valid will, under which everyone can be free.

Utilitarian Ethics

The Utilitarian ethics was proposed by John Stuart. According to this theory, the happiness or pleasure of a greatest number of people in the society is considered as the greatest good. According to this philosophy, an action is morally right if its consequences lead to happiness of people and wrong if they lead to their unhappiness.

An example of this can be the removal of reservation system in education and government jobs, which can really benefit the talented. But this can have an impact on the rights of the minorities.

Problem of Application

Qualification of the benefits can be difficult.

What are these ethics?

Consider the cost-benefit analysis in engineering. A typical cost-benefit analysis identifies the good and bad consequences of some action or policy in a monetary aspect. It weighs the total good against total bad and then compares the results to similar tallies of the consequences of alternative actions or rules. This supports the idea of maximizing benefits against cost.

There are two main types of Utilitarianism. They are –

Act Utilitarianism

The Act Utilitarianism focuses on each situation and the alternative actions possible in the situation. Act Utilitarianism states that A particular action is right if it is likely to produce the higher level of good for the most people in a given situation, compared to alternative choices that might be made.

In accordance with this theory, the good done is only considered but not the way how it is done. For example, looting the richer to feed the poor, can satisfy and make a group of poor people, happy. But looting is not a way of morality. Hence act-utilitarianism seems to justify the wrong-doing.

Rule Utilitarianism

The Rule Utilitarianism states that Right actions are those required by rules that produce the higher level of good for the most people. We need to consider a set of rules, where they interact with each other. This was developed to clear the problem that occurs with act-utilitarianism.

Engineers with ethics should follow the rule-utilitarianism considering the point, Act as faithful agents or trustees of employers. So, engineers should abide by it even when an exception might happen to be beneficial. Like in the above example, one should seek the help of law and order to prove the guilt of richer and let see that the poor get benefitted.

Formulation of Ethical Theories

After having gone through the various ethical theories, one can understand that these ethical theories have to be formulated considering the following points –

- The concepts of the theory formulated must be coherent.
- The tenets of the theory should never contradict the other.
- The theory should never be defended upon false information.
- The theory should guide in specific situations comprehending all aspects possible.
- The theory should be compatible with individuals moral convictions in any situation.

Uses of Ethical Theories

Ethical theories help in the following areas –

- Understanding moral dilemmas.
- Justifying professional obligations and ideas.
- Relating ordinary and professional morality.

Engineering Ethics - Moral Issues

A moral issue can be understood as an issue to be resolved not only by considering the technical stuff but also by keeping moral values in mind. To be more precise, let us consider the definition in general.

Moral issue is a working definition of an issue of moral concern is presented as any issue with the potential to help or harm anyone, including oneself.

Types of Moral Issues

There are mainly two types of Moral issues that we mostly come across while keeping the ethical aspects in mind to respond. They are –

Micro-ethics

This approach stresses more on the problems that occur on a daily basis in the field of engineering and its practice by engineers.

Macro-ethics

This approach deals with social problems which are unknown. However, these problems may unexpectedly face the heat at both regional and national levels.

Examples

Let us now understand a few examples related to moral issues.

Example 1

After a recent collapse of a structure in which many people died, an Engineer came to know about a bridge which is marginally safe. He informed his superior who asked him to stay calm and not to discuss with anyone, while waiting for the next year budget sessions to get some financial help for the repair required. What should the engineer do?

Example 2

What should an Engineer who observes his colleague copying confidential information unauthorized, do immediately? If he chooses to stop his friend, what if this gets repeated without his notice? If he chooses to report the management, what if his friend loses the job? Which is morally correct?

Example 3

An engineer who develops a proto-type for the project, loses it due to a mishap exactly the day before the submission. Is it morally correct to outsource the prototype of the project and reduce the risks of job insecurity? What should he do?

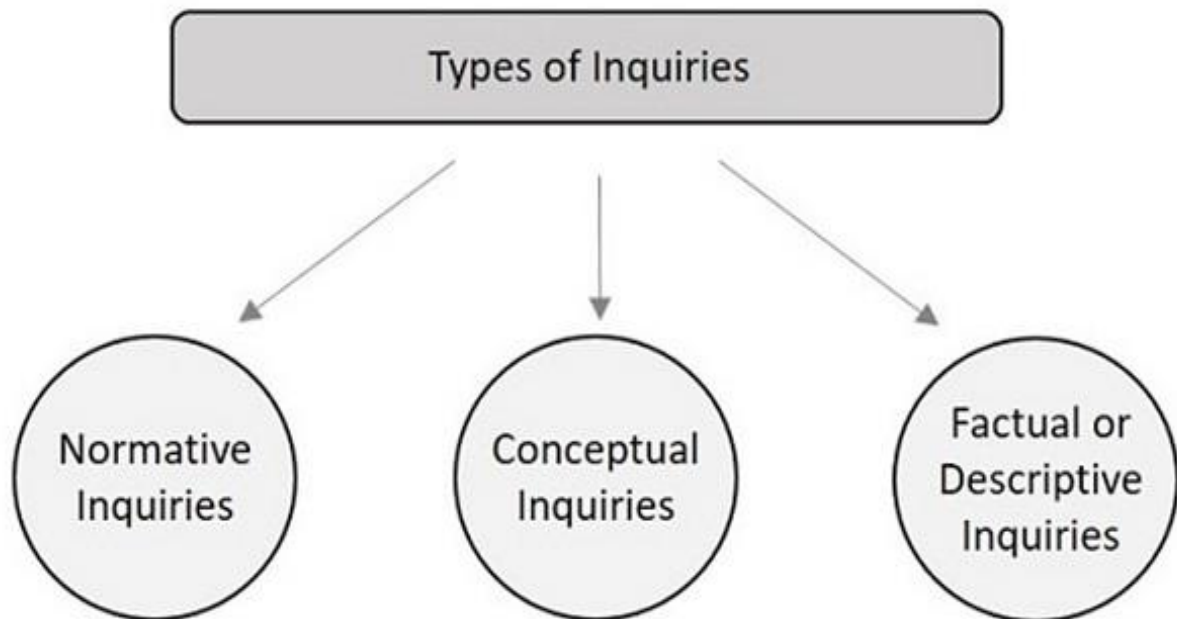
These are the few examples just to understand the kind of moral dilemmas. There might be one or more correct answers at times. There can be some other way around to deal with the issue, which one cant easily notice. However, the decisions have to be made by following a slow and clear process in order to avoid further problems and also to solve this in a manner that leads to no regrets.

Types of Inquiries

The issues can be resolved by following an investigation procedure, step by step in order to have a clear understanding towards the issue. Here we have three different types of inquiries.

Judging the issues has to be followed by a systematic procedure to avoid any flaws. Engineering ethics involves investigations into values, meanings and facts. Following are the different types of inquiries made for this.

- Normative inquiries
- Conceptual inquiries
- Factual or descriptive inquiries



Normative Inquiries

Normative Inquiry refers to the description that describes **what one ought to do** under a specific circumstance. This is the expected ideal response, which might differ from what one believes to be right or wrong.

This list identifies and justifies the morally desirable nature for guiding individuals or groups. This includes the responsibility of engineers to protect the public safety and how they should respond under such dangerous practices. Normative inquiries also quote the laws and procedures that affect the engineering practice on moral grounds. They refer to the thought process where the moral rights are to be implemented in order to fulfill their professional obligations.

Conceptual Inquiries

Conceptual Inquiry refers to the description of the meaning of concepts, principles and issues related to engineering ethics. The ethics that an engineer should possess to protect the safety, health and welfare of the public, etc. are described under conceptual inquiries.

It describes what safety is and mentions the marginal issues of safety along with the precautions an engineer should take to avoid risk. Conceptual inquiries mention the moral aspects of bribery and how its effects, along with the professional ethics and professionalism.

Factual and Descriptive Inquiries

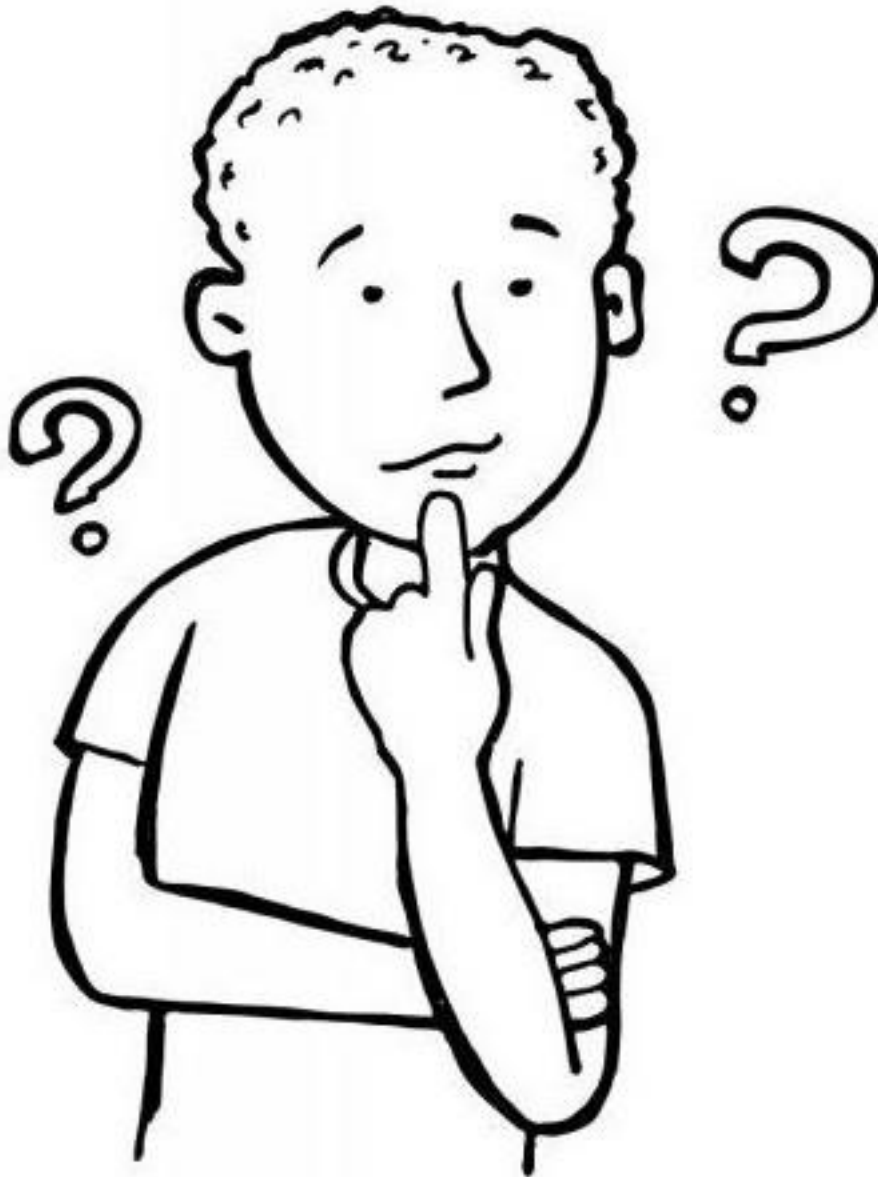
Factual Inquiry or the descriptive inquiry help to provide the facts for understanding and finding solutions to the value based issues. The engineer has to conduct factual inquiries by using scientific techniques.

This helps in providing the information regarding the business realities such as engineering practice, history of engineering profession, the effectiveness of professional societies, the procedures to be adopted when assessing risks and psychological profiles of engineers.

Let us now go through the concept of Moral dilemma that a person faces when confronted with a situation.

Engineering Ethics - Moral Dilemmas

At times, the situations occur where one cannot make immediate decisions as the moral reasons come into conflict. The moral reasons can be rights, duties, goods or obligations, which make the decision making complex.



Types of Complexities

The difficulties in arriving to a solution, when segregated, can be divided into the following three sections.

Vagueness

This refers to the condition where the doubt lies in whether the action refers to good or bad. This is just like having a thought that following the rules is mandatory. This sometimes includes the unwritten rules like being loyal, having respect, maintaining confidentiality, etc.

Conflicting reasons

When you know about the solutions you have, the making of better choice among the ones you have, will be the internal conflict. Fixing the priorities depends upon the knowledge and the moral values one has. The reason why the particular choice is being made, makes sense.

Disagreement

When there are two or more solutions and none among them is mandatory, the final solution selected should be best suitable under existing and the most probable conditions. The interpretation regarding the moral reasons behind the choice and analysis should be made keeping in mind whether this is the better or the worse solution in the probable aspects.

Steps in Facing Moral Dilemmas

Whenever a person is faced with a moral dilemma, the issue is to be solved with a stepwise approach as this will generate a better output. The steps include the following –

Identification

The step of identification involves the following –

- The issue has to be thoroughly understood.
- The duties and the responsibilities of the persons involved are to be clearly known.
- The moral factors related to the issue are to be understood.
- The conflicting responsibilities, the competing rights and the clashing ideas involved are to be identified.

Ranking

The considerations in the issue are to be listed down. Then they have to be ranked according to the priorities. The moral aspect has to be considered to rank the issues. The advantages of a single person should never be given any importance unless any moral reason is there behind it. No partiality is allowed.

Inquiries

The inquiry of details involved in the issue is to be completely made. All the facts related to the issue are brought into light. Considering the alternative courses of action for resolving and tracing, full implications are also needed.

Discussions

Discussions are to be made with other members, as different minds look at the issue in different views to give different solutions. The complete analysis of a problem gives chances to different viewpoints, perspectives and opinions from which a better solution can be drawn.

Final Solution

After analyzing different perspectives and considering the facts and reasons on the basis of truths and understanding the flaws which lead to the issue, a final solution has to be drawn out. This solution will add value to the whole analysis, in all aspects.

Engineering Ethics - Moral Autonomy

Moral Autonomy is the philosophy which is self-governing or self-determining, i.e., **acting independently** without the influence or distortion of others. The moral autonomy relates to the individual ideas whether right or wrong conduct which is independent of ethical issues. The concept of moral autonomy helps in improving self-determination.

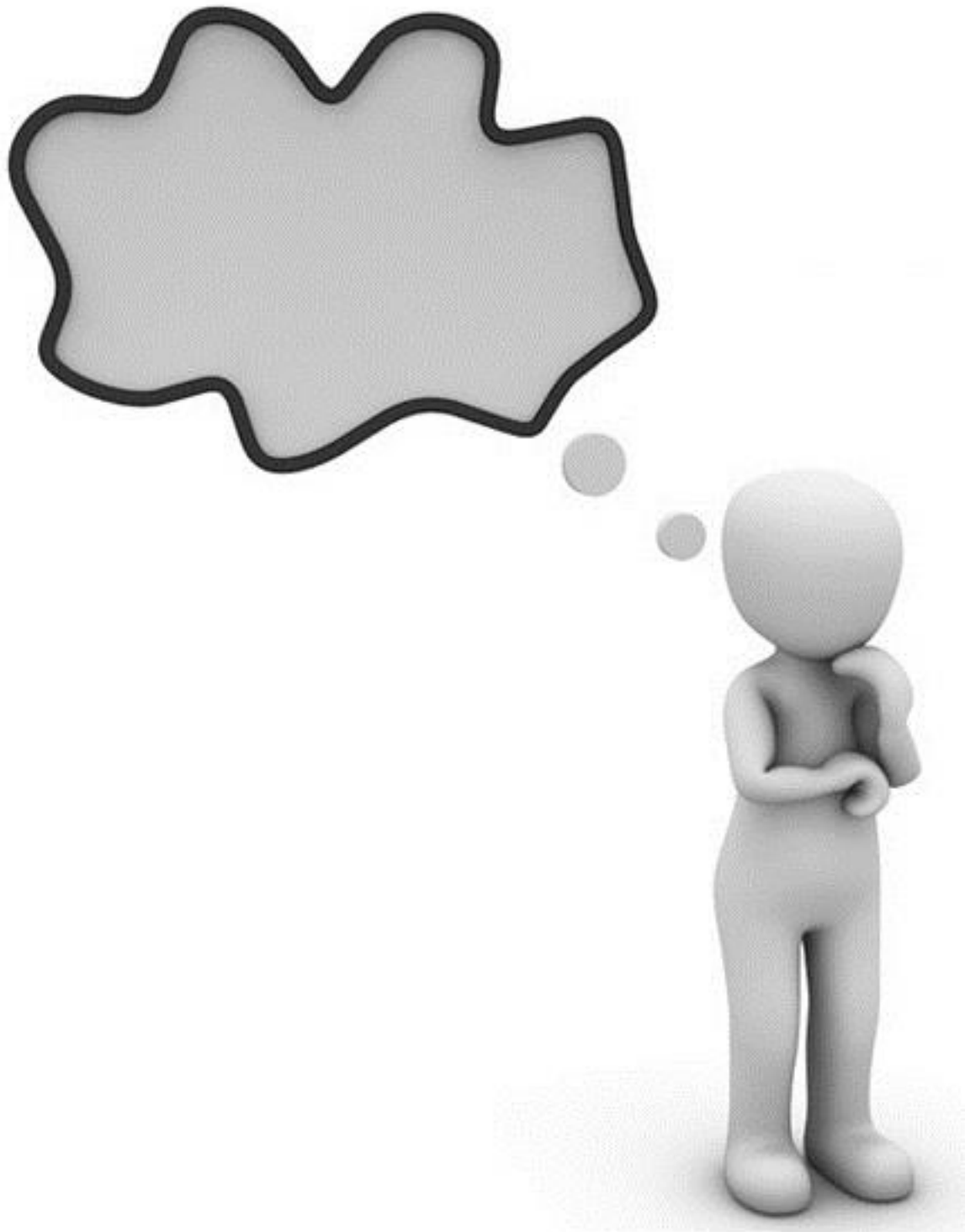
Moral Autonomy is concerned with independent attitude of a person related to moral/ethical issues. This concept is found in moral, ethical and even in political philosophy.

Moral Autonomy Skills Needed

In this section, let us discuss the skills needed for moral autonomy.

- **Ability to relate the problems with the problems of law, economics and religious principles** – It is essential to have the ability to analyze a problem and finding the relation with the existing law or the topic of issue with the existing principles on that topic. The ability to distinguish between both of them and finding the moral reasons.
- **Skill to process, clarify and understand the arguments against the moral issues** – If the issue is against some moral values or the ethical values to be followed in the society, then clarity should be maintained about the differences and similarities. Both of these differences and similarities are to be judged based on why they are a matter of concern and in what aspect.
- **Ability to suggest the solutions to moral issues on the basis of facts** – If the moral issues are not fulfilling and needs to be, then the solutions are to be suggested according to the moral issues based on the facts and truths of the issue. These suggestions must be consistent and must include all the aspects of the problem. No partiality is to be allowed in any such aspect.
- **Must have the imaginative skill to view the problems from all the viewpoints** – After having known about the facts and illusions of the issue, a clear understanding is attained in viewing the problem in all kinds of viewpoints. This enables one to be able to suggest a proper alternative solution.
- **Tolerance while giving moral judgment, which may cause trouble** – When the whole analysis is made considering all the viewpoints of the issue, the final output might be or might not be pleasing to the persons involved. Hence while declaring the judgment or the decisions taken, a detailed description of the actions done should be given, while the actions ought to be done should be presented in a better way, to ensure others that the decisions have been taken without any partialities towards any party.
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Skills for Improving Moral Autonomy

Moral autonomy reflects the concept of individuality. This relates to the idea of building ones self with the moral values one has while developing psychologically.

To have moral autonomy in all the aspects, one should have a lot of **patience** and interest. One should adhere to the basic principles of humanity and should be strict with the Dongs he has in mind and liberal with his Dos. The kindness towards his fellow beings is also an important concept to be kept in mind. Inculcation of all these important qualities, enhances the skills of Moral autonomy in a person.

A Person must have adequate knowledge and understanding about the use of ethical language so as to defend or support his views with others. He must have better **knowledge** in understanding the importance of suggestions and better solutions while resolving moral problems and also about the importance of tolerance on some critical situations.

Above all, one must understand the importance of maintaining **moral honesty** and should be liberal to understand the human behavior under certain circumstances.